

Focal Nodular Hyperplasia and Focal Nodular Hyperplasia-like Lesions

Jordan D. LeGout, MD
 Candice W. Bolan, MD
 Andrew W. Bowman, MD, PhD
 Melanie P. Caserta, MD
 Frank K. Chen, MD
 Kelly L. Cox, MD
 Rupan Sanyal, MD
 Beau B. Toskich, MD
 Jason T. Lewis, MD
 Lauren F. Alexander, MD

Abbreviations: APHE = arterial phase hyperenhancement, BCS = Budd-Chiari syndrome, FALD = Fontan-associated liver disease, FNH = focal nodular hyperplasia, HCC = hepatocellular carcinoma, HBP = hepatobiliary phase, HCA = hepatocellular adenoma, HHT = hemorrhagic hereditary telangiectasia, HSCA = hepatocyte-specific contrast agent, LI-RADS = Liver Imaging Reporting and Data System, NRH = nodular regenerative hyperplasia

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From the Department of Radiology (J.D.L., C.W.B., A.W.B., M.P.C., F.K.C., K.L.C., R.S., B.B.T., L.F.A.) and Department of Laboratory Medicine and Pathology (J.T.L.), Mayo Clinic Florida, 4500 San Pablo Rd, Jacksonville, FL 32224. Presented as an education exhibit at the 2020 RSNA Annual Meeting. Received May 12, 2021; revision requested June 17 and received July 18; accepted July 20. For this journal-based SA-CME activity, the authors, editor, and reviewers have disclosed no relevant relationships. Address correspondence to J.D.L. (e-mail: LeGout.Jordan@mayo.edu).

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Recognize the typical and atypical imaging features of FNH and FNH-like lesions at CT, US, and MRI.
- Differentiate FNH and FNH-like lesions from other liver lesions at imaging.
- Discuss the various conditions that predispose a patient to formation of FNH-like lesions and recognize their main imaging features.

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Focal nodular hyperplasia (FNH) is a benign lesion occurring in a background of normal liver. FNH is seen most commonly in young women and can often be accurately diagnosed at imaging, including CT, MRI, or contrast-enhanced US. In the normal liver, FNH frequently must be differentiated from hepatocellular adenoma, which although benign, is managed differently because of the risks of hemorrhage and malignant transformation. When lesions that are histologically identical to FNH occur in a background of abnormal liver, they are termed FNH-like lesions. These lesions can be a source of diagnostic confusion and must be differentiated from malignancies. Radiologists' familiarity with the imaging appearance of FNH-like lesions and knowledge of the conditions that predispose a patient to their formation are critical to minimizing the risks of unnecessary intervention for these lesions, which are rarely symptomatic and carry no risk for malignant transformation. FNH is thought to form secondary to an underlying vascular disturbance, a theory supported by the predilection for formation of FNH-like lesions in patients with a variety of hepatic vascular abnormalities. These include abnormalities of hepatic outflow such as Budd-Chiari syndrome, abnormalities of hepatic inflow such as congenital absence of the portal vein, and hepatic microvascular disturbances, such as those that occur after exposure to certain chemotherapeutic agents. Familiarity with the imaging appearances of these varied conditions and knowledge of their association with formation of FNH-like lesions allow radiologists to identify with confidence these benign lesions that require no intervention.

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Introduction

Focal nodular hyperplasia (FNH) is a benign lesion of hepatocytic hyperplasia that arises in a background of normal or nearly normal liver. Histologically and radiologically identical lesions in a background of an abnormal liver are called *FNH-like lesions*. Although FNH is most common in young women, FNH-like lesions are seen in myriad populations with liver abnormalities who share an underlying cause of hepatic vascular disturbance, including those with abnormalities of hepatic inflow and outflow and hepatic microvascular disturbances (Table 1). Familiarity with the imaging features of FNH and FNH-like lesions and the various predisposing conditions can allow confident diagnosis at imaging to avoid undue patient anxiety and the risks of biopsy or surgery.

Focal Nodular Hyperplasia

FNH is the second most common benign liver lesion after hemangioma, representing 8% of primary hepatic lesions (1). It has an incidence of 0.9%, is much more common in women than in men

TEACHING POINTS

- Four patterns of HBP contrast material uptake have been described for FNH: homogeneous hyperintensity, heterogeneous hyperintensity, homogeneous isointensity, and hypointensity with peripheral ring uptake.
- If a lesion demonstrates typical MRI characteristics of FNH during all sequences, then a confident diagnosis can be made. However, if all classic features are not present or one or more atypical features are identified, then follow-up imaging or tissue diagnosis should be pursued on the basis of the degree of suspicion for a non-FNH lesion.
- *FNH-like lesion* is the term applied to lesions that are macroscopically, microscopically, and immunohistochemically identical to FNH but occur in a background of abnormal liver. Myriad liver conditions predispose patients to formation of FNH-like lesions, all of which share a vascular disturbance as the inciting cause.
- The Liver Imaging Reporting and Data System (LI-RADS) acknowledges this decreased specificity for imaging diagnosis of HCC in the populations who are predisposed to formation of FNH-like lesions and explicitly states that LI-RADS criteria should not be applied in patients with cardiogenic cirrhosis (which includes FALD) or with cirrhosis secondary to vascular disorders, including hereditary hemorrhagic telangiectasia (HHT), Budd-Chiari syndrome (BCS), chronic portal vein occlusion, and nodular regenerative hyperplasia (NRH).
- Metastases from the adult malignancies that are commonly treated with oxaliplatin regimens are most commonly hypoenhancing at MRI. Given the possibility of development of FNH-like lesions in these patients, the determination that a new arterially enhancing lesion at follow-up imaging is a metastasis should be made with caution.

(8:1), and is most commonly diagnosed in the 3rd to 5th decades of life. FNH is usually a singular lesion, but there may be multiple lesions in 20% of cases (2).

Histologically, FNH is hyperplastic growth of morphologically normal hepatocytes, without normal development of the portal tract. A central fibrovascular scar and radiating fibrous septa contain large malformed feeder arteries and branches. Ductular proliferation is characteristically present along the septal edges (2). Although the pathogenesis is not fully understood, most researchers agree that FNH forms as a hyperplastic response to abnormal hepatic blood flow (3).

Most cases of FNH are asymptomatic and incidental. FNH lesions have no malignant potential, and hemorrhage and rupture are rare, with only 10 reported cases in the literature (4). Consequently, asymptomatic FNH requires no treatment or imaging follow-up in patients with no known malignancy or underlying liver disease (5). FNH is rarely symptomatic when the lesion is large or secondary to mass effect. Surgery or local-regional therapy, most commonly transarterial bland embolization, are both effective for management of symptomatic FNH (6).

Table 1: Conditions Predisposing Patients to Formation of FNH-like Lesions

Abnormalities of hepatic outflow
Budd-Chiari syndrome
Fontan-associated liver disease
Other forms of cardiogenic congestive hepatopathy
Abnormalities of hepatic inflow
Congenital absence of the portal vein
Cavernous transformation
Hereditary hemorrhagic telangiectasia
Hepatic microvascular disturbances
Cirrhosis
Nodular regenerative hyperplasia
Chemotherapy induced

Typical Imaging Features of FNH

General Imaging Features.—The imaging features of FNH correspond well to the histologic features and are similar across modalities. FNH is typically a solitary well-circumscribed, unencapsulated, lobulated mass that usually measures smaller than 5 cm (7). Because FNH is mainly composed of hepatocytes, it appears similar to the background liver on unenhanced images. Accurate diagnosis of FNH at routine US is only possible in one-third of cases, with subtle, if any, difference in echogenicity when compared with the surrounding liver (8). Color and power Doppler US improve visualization by demonstrating the central feeding artery and characteristic “spoke-wheel” pattern of branching vessels in the radiating septa (9,10) (Fig E1). Similarly, FNH lesions are isoattenuating to slightly hypoattenuating on nonenhanced CT images and may be isointense or mildly T1-hypointense or slightly T2-hyperintense on MR images, thus earning the term *stealth lesion* (7).

The large feeding artery and central branching vessels explain the characteristic homogeneous arterial phase hyperenhancement (APHE) seen at contrast-enhanced US, CT, and MRI (7). At contrast-enhanced US, the APHE is in a characteristic centrifugal pattern with spoke-wheel enhancement or an eccentric feeding vessel (9) (Fig 1). Contrast-enhanced US has diagnostic accuracy of 87% for FNH, and continuous cine imaging is more sensitive than other modalities for detecting the early arterial spoke-wheel sign (10,11). In the portal and later phases, FNH remains slightly hyperenhancing or becomes isoenhancing to the adjacent liver with all modalities (7,9) (Figs 2, 3).

Histologically, a central scar composed of fibrous and myxomatous elements is present in

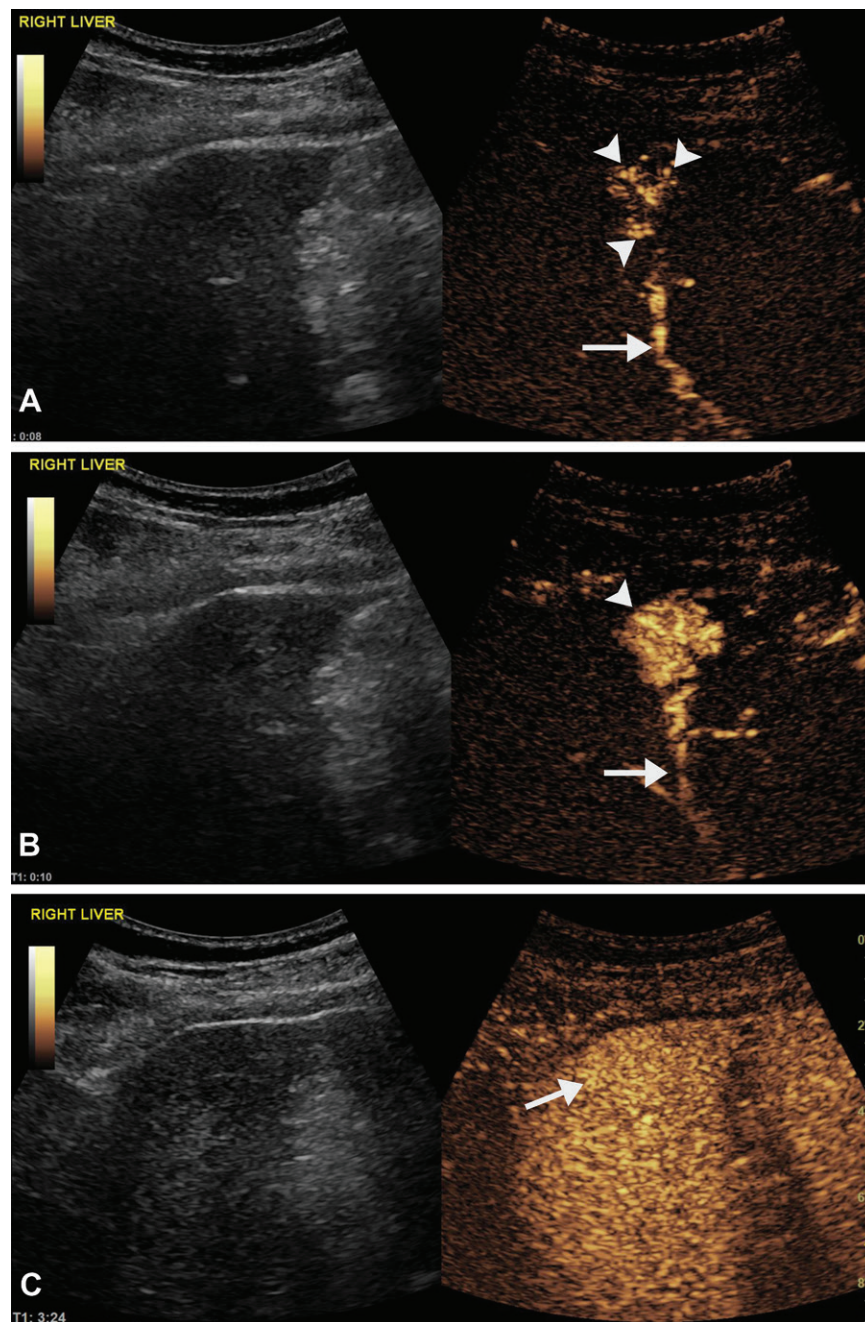


Figure 1. Focal nodular hyperplasia (FNH) in a 70-year-old woman. (A) Dual-screen contrast-enhanced US images at 8 seconds after administration of a microbubble contrast agent (sulfur hexafluoride lipid-type A microspheres) show the characteristic large feeding artery (arrow) and spoke-wheel enhancement pattern (arrowheads). (B) Dual-screen contrast-enhanced US images at 10 seconds show the hyperenhancing mass with the large feeding artery (arrow), and spoke-wheel enhancement (arrowhead) is still visible. (C) Dual-screen contrast-enhanced US images at 3 minutes and 24 seconds show that the lesion is nearly isoenhancing in comparison with the liver (arrow).

almost all cases of FNH but is visible in only approximately 60% of lesions with CT and in 80% of lesions with MRI (7). A large feeding artery may be visualized in the central scar but is best depicted on early angiographic images and may not be identified during the late hepatic arterial phase, when images of the liver are routinely acquired (Fig 2A). At CT and MRI with extracellular contrast agents, the scar is hypoenhancing on arterial and portal venous phase images and shows delayed hyperenhancement at 2–5 minutes (7) (Figs 2, 3).

MRI-specific Features.—MRI is generally considered the most sensitive and specific modality for

detection and diagnosis of FNH, particularly with hepatocyte-specific contrast agents (HSCAs).

With diffusion-weighted MRI, most FNH lesions are mildly hyperintense in comparison with the normal liver on both low- and high-*b*-weighted images (12). Although FNH lesions generally have higher apparent diffusion coefficients than those of malignant lesions and hepatocellular adenomas (HCAs), substantial overlap in apparent diffusion coefficients exists, and characteristics at diffusion-weighted MRI are not a reliable differentiating feature in isolation (13). At MRI, the central scar is characteristically T2 hyperintense, T1 hypointense, and hyperintense on apparent diffusion coefficient maps (8) (Fig 3).

HSCAs provide additional diagnostic information and have dual functions, acting as extracellular contrast agents up to the venous phase and transitioning to HSCA uptake over time. Hepatobiliary phase (HBP) imaging is typically performed at 20 minutes after administration of gadoxetate disodium (Eovist, Bayer Healthcare) or 120 minutes after administration of gadobenate dimeglumine (MultiHance, Bracco). HSCA uptake and excretion into bile is facilitated by hepatocyte-specific membrane transport proteins that are absent in other cell types (14). FNH, as a hyperplastic growth of hepatocytes, almost always shows HBP uptake, whereas lesions of nonhepatocellular origin, such as hemangiomas or metastases, do not take up HSCAs and appear hypointense during the HBP. Other lesions of hepatocellular origin such as hepatocellular carcinoma (HCC) and HCA are also usually hypointense during the HBP; however, these lesions may retain or have up-regulated hepatocyte-specific membrane transport proteins, and therefore can be isointense or even hyperintense on images acquired during the HBP, in a minority of cases (15).

The arterial and portal venous phase enhancement pattern of FNH with HSCAs and extracellular contrast agents is identical. However, the central scar typically remains hypointense with gadoxetate enhancement as it begins to transition from extracellular characteristics to hepatocyte-specific uptake before the time frame typical for delayed enhancement of the central scar (16) (Fig 4).

Four patterns of HBP uptake of HSCA have been described for FNH: homogeneous hyperintensity (Fig 4), heterogeneous hyperintensity, homogeneous isointensity, and hypointensity with peripheral ring uptake (17,18) (Fig 5). Most cases of FNH demonstrate HSCA uptake, with only 2% reported as hypointense during the HBP (18). It is important to differentiate typical FNH HBP uptake patterns from atypical patterns of HBP enhancement that are associated with malignancies (Table 2). A targetoid pattern during the HBP refers to central contrast enhancement with a peripheral hypointense rim, which is secondary to HSCA retention in tumors with fibrous stroma rather than to cellular uptake. This pattern is associated with cholangiocarcinoma and metastatic adenocarcinoma. One must also differentiate the peripheral ring uptake pattern of FNH (in which the peripheral uptake is in the FNH) from peritumoral retention, where the hyperintense uptake is in a peripheral rim of reactive peritumoral hepatocytic hyperplasia surrounding the lesion and not within the lesion itself, which is a pattern that

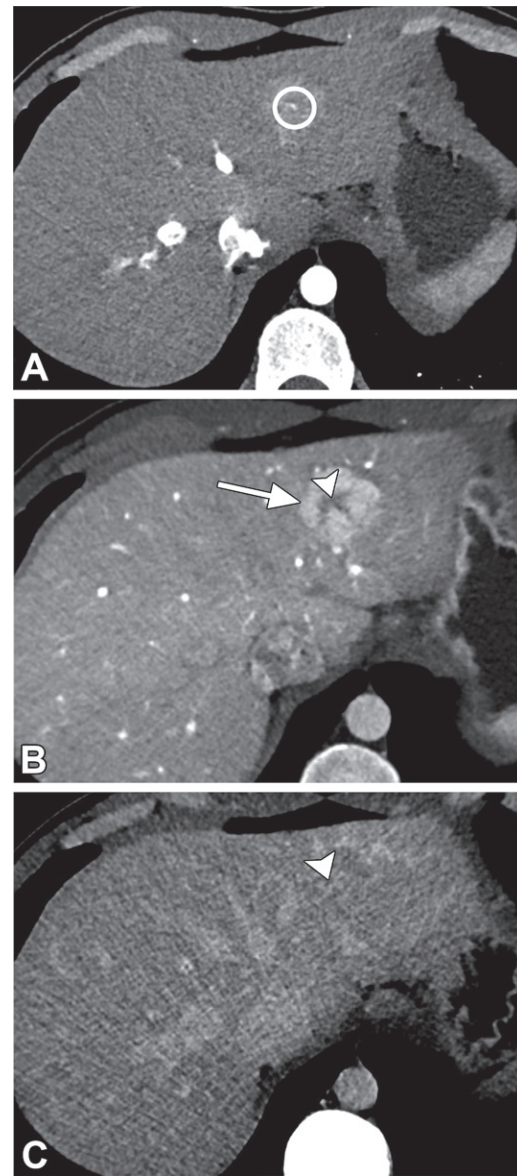


Figure 2. Typical CT appearance of FNH in a 29-year-old man undergoing evaluation as a potential kidney donor. (A) Axial early arterial phase CT image shows a large feeding artery (circle) in the central scar. (B) Axial late arterial phase CT image shows homogeneous hyperenhancement of the FNH lesion (arrow), with a hypoattenuating central scar (arrowhead). (C) Axial delayed phase CT image shows that the mass is isoenhancing with a hyperenhancing central scar (arrowhead).

is associated with a variety of non-FNH lesions. Excreted HSCAs in intralesional or perilesional bile ducts should not be mistaken for hepatocytic contrast material uptake (19) (Fig 6).

The main conditions to consider in the differential diagnosis of a liver lesion with APHE in a young patient with a normal liver and no malignancy or risk factors for liver disease are FNH and HCA. HCA is a rare neoplasm that is most common in young women and carries substantial risks for malignant transformation (4%) and hem-

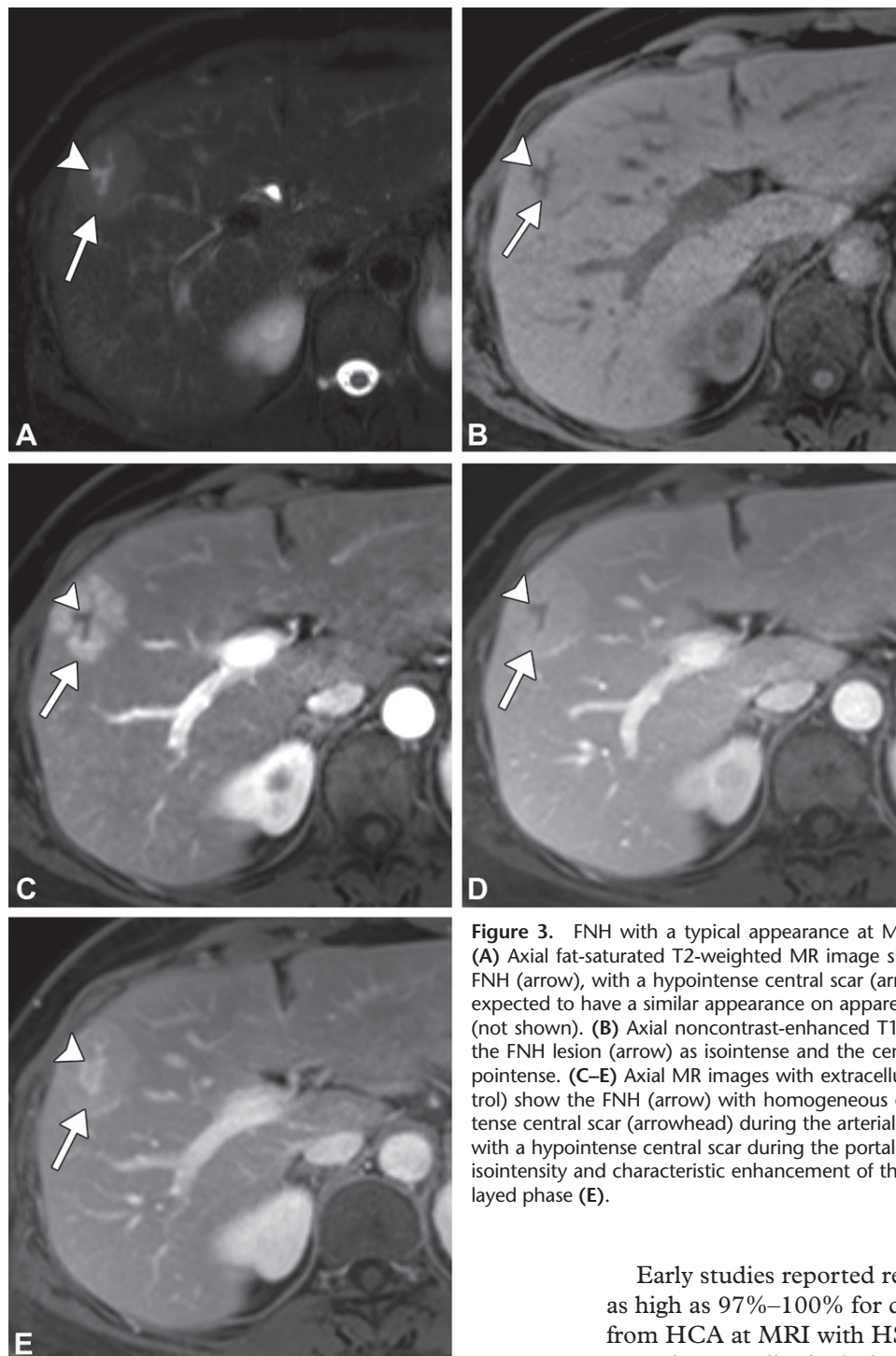


Figure 3. FNH with a typical appearance at MRI in a 46-year-old woman. (A) Axial fat-saturated T2-weighted MR image shows a slightly hyperintense FNH (arrow), with a hypointense central scar (arrowhead). The central scar is expected to have a similar appearance on apparent diffusion coefficient maps (not shown). (B) Axial noncontrast-enhanced T1-weighted MR image shows the FNH lesion (arrow) as isointense and the central scar (arrowhead) as hypointense. (C–E) Axial MR images with extracellular contrast agent (gadobutrol) show the FNH (arrow) with homogeneous enhancement and a hypointense central scar (arrowhead) during the arterial phase (C); near isointensity, with a hypointense central scar during the portal venous phase (D); and near isointensity and characteristic enhancement of the central scar during the delayed phase (E).

orrhage (16%) (20,21). HCA represents a diverse class of tumors that for the purposes of imaging can be considered one of four subtypes: HNF1- α inactivated, inflammatory, β -catenin activated, and unclassified (22). Accurate differentiation of FNH and HCA is critical because of the stark differences in their management (Table E1). Given the risks, HCA requires at least imaging surveillance, if not biopsy for characterization of the subtype, stratification of risk, and planning of treatment.

Early studies reported remarkable accuracy as high as 97%–100% for differentiating FNH from HCA at MRI with HSCAs (23). However, these studies included a limited number of lesions, lacked or included only partial HCA subtyping, and may have allowed misclassification of some inflammatory HCAs as telangiectatic FNH. Subsequent studies have revealed lower accuracy rates of 87%–95% (22,24). Although most (75%–90%) HCAs are hypointense during the HBP, approximately 25% of inflammatory and 40%–80% of β -catenin-activated HCAs are reported to show isointense or hyperintense uptake during the HBP, mimicking FNH (22,25) (Fig 7A). This overlap with FNH is problematic, because β -catenin HCAs, although rare, have the

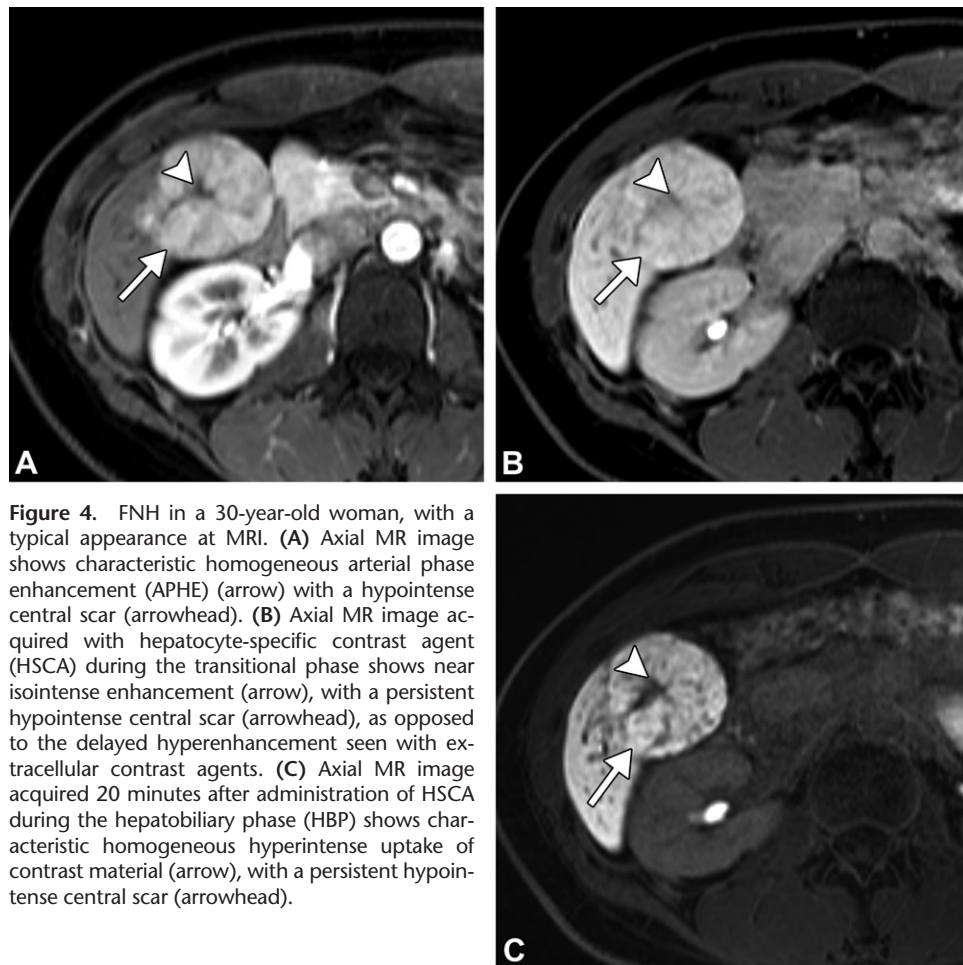


Figure 4. FNH in a 30-year-old woman, with a typical appearance at MRI. (A) Axial MR image shows characteristic homogeneous arterial phase enhancement (APHE) (arrow) with a hypointense central scar (arrowhead). (B) Axial MR image acquired with hepatocyte-specific contrast agent (HSCA) during the transitional phase shows near isointense enhancement (arrow), with a persistent hypointense central scar (arrowhead), as opposed to the delayed hyperenhancement seen with extracellular contrast agents. (C) Axial MR image acquired 20 minutes after administration of HSCA during the hepatobiliary phase (HBP) shows characteristic homogeneous hyperintense uptake of contrast material (arrow), with a persistent hypointense central scar (arrowhead).

highest risk for malignant transformation (40%) and generally require surgical resection (26). Furthermore, HCAs typically demonstrate APHE and can mimic FNH in other ways, including the presence of a central scar in 43% of inflammatory and 67% of β -catenin HCAs in one study (25), which adds to the diagnostic dilemma (Fig 7B).

In addition, studies of MRI with HSCAs for differentiating among a larger variety of liver lesions in noncirrhotic livers have shown even lower accuracy, (66%–79% in a study including FNH, HCA, and HCC and 64%–79% in a study with a variety of focal liver lesions) (27,28). These results suggest that the value of imaging during the HBP to differentiate among FNH, HCA, and other focal liver lesions may be overemphasized, and caution is warranted against an overreliance on the appearance during the HBP as a single defining feature. Although HSCAs remain an important assessment tool, the most accurate characterization requires scrutiny of all imaging features. If a lesion demonstrates typical MRI characteristics of FNH during all sequences, then a confident diagnosis can be made. However, if all classic features are not present or one or more atypical features are identified, then follow-up imaging or tissue diagnosis should be pursued on the basis of the degree of suspicion for a non-FNH lesion.

Atypical Imaging Features of FNH

Most FNH lesions show typical imaging features, allowing confident diagnosis. However, it is important to be familiar with atypical manifestations and their relative frequency to direct patient treatment most accurately (Table E2).

Most (85%) FNH lesions are smaller than 5 cm, and only a small percentage are larger than 10 cm (29) (Fig E2). Although they were originally described as static, it is now known that 12%–15% of FNH lesions grow (Fig E3) and approximately 9% regress over time (30,31) (Fig E4). Growth may cause clinical uncertainty, but is not associated with patient symptoms or other adverse outcomes and should not be considered an indication for surgical resection; however, referral to a tertiary care center could be considered to confirm the accuracy of the imaging diagnosis and to facilitate multidisciplinary management (30,31).

Intralesional steatosis is commonly seen histologically in FNH but it is rarely seen at imaging (2). Intralesional fat, which is identified as areas of loss of signal intensity at chemical shift MRI, is much more commonly associated with HCAs and HCCs, so this finding raises diagnostic uncertainty when it is present in a patient with FNH. Limited

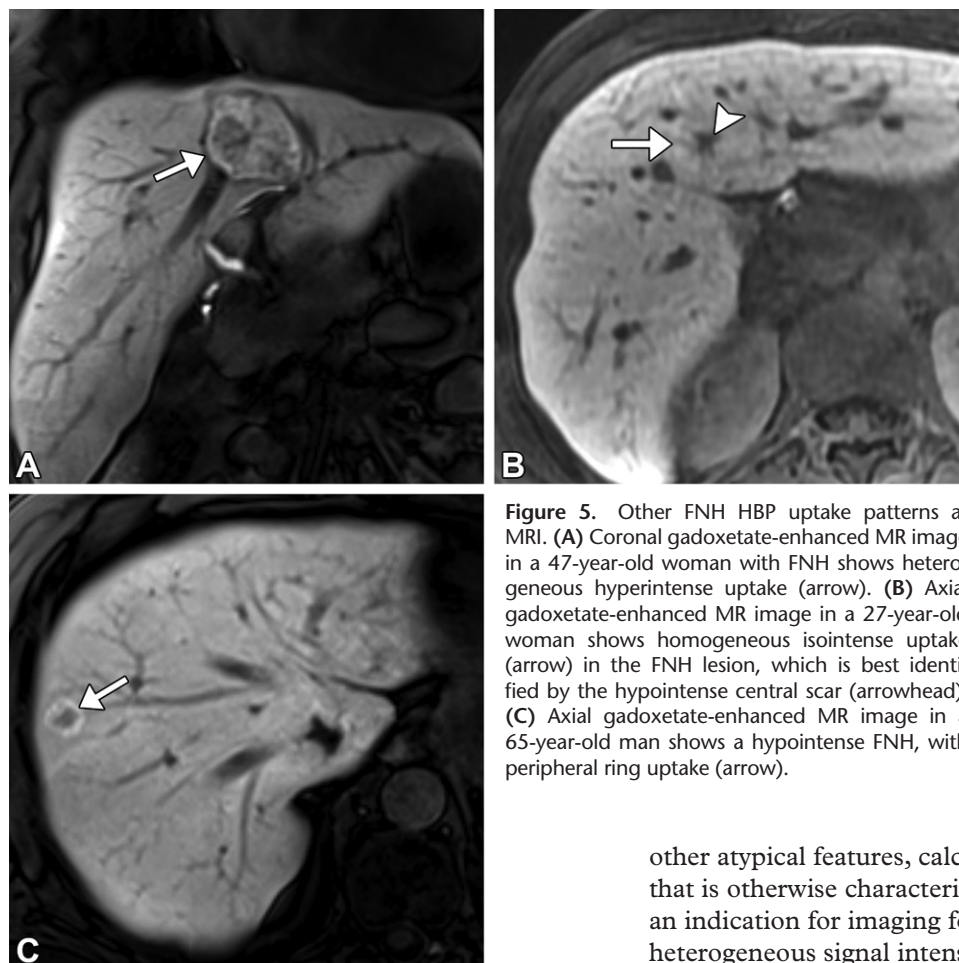


Figure 5. Other FNH HBP uptake patterns at MRI. (A) Coronal gadoxetate-enhanced MR image in a 47-year-old woman with FNH shows heterogeneous hyperintense uptake (arrow). (B) Axial gadoxetate-enhanced MR image in a 27-year-old woman shows homogeneous isointense uptake (arrow) in the FNH lesion, which is best identified by the hypointense central scar (arrowhead). (C) Axial gadoxetate-enhanced MR image in a 65-year-old man shows a hypointense FNH, with peripheral ring uptake (arrow).

Table 2: HBP Patterns of Contrast Material Uptake of FNH and Non-FNH Lesions

HBP uptake patterns of FNH lesions

- Homogeneous hyperintense
- Heterogeneous hyperintense
- Homogeneous isointense
- Hypointense with peripheral ring uptake

HBP patterns of non-FNH lesions

- Targetoid (central uptake with peripheral hypointensity)
- Peritumoral hyperintense rim
- Excretion of contrast in peri- or intralesional bile ducts

data suggest that if all other classic features of FNH are present, then a confident diagnosis of FNH with intralesional fat can still be made (32) (Fig 8). In isolation, this feature is an indication for imaging follow-up. If at least one of the other typical features of FNH is absent, then biopsy should be considered.

Calcification in patients with FNH is rare, occurring in approximately 1% of lesions, typically in the central scar (33) (Fig 9). As with

other atypical features, calcification in a lesion that is otherwise characteristic of FNH may be an indication for imaging follow-up. Marked or heterogeneous signal intensity on T2-weighted MR images and atypical enhancement patterns including the absence of APHE (4%) and signal hypointensity on HBP (2%) MR images are rare for patients with FNH but make correct diagnosis with imaging difficult (7,18). Tissue diagnosis is likely required for this subset of lesions.

FNH-like Lesions

FNH-like lesion is the term applied to lesions that are macroscopically, microscopically, and immunohistochemically identical to FNH but occur in a background of abnormal liver (Fig E5). Myriad liver conditions predispose patients to formation of FNH-like lesions, all of which share a vascular disturbance as the inciting cause (Table 1) (34). Because they are histologically identical to FNH, FNH-like lesions carry no risk for malignant degeneration. Although they may create diagnostic confusion, they are otherwise typically clinically silent.

Abnormalities of Hepatic Outflow

Fontan-associated Liver Disease.—The Fontan circulation procedure, a common palliative repair technique for patients with complex congenital heart disease, allows blood from systemic venous return to bypass the heart and enter the

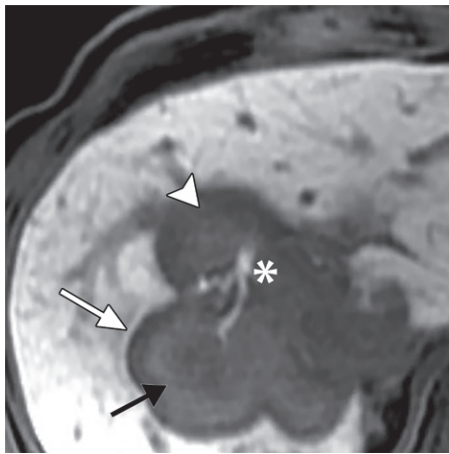


Figure 6. Intrahepatic cholangiocarcinoma in a 54-year-old man. Axial gadoxetate-enhanced HBP MR image shows the targetoid appearance of the mass, with areas of central contrast material retention (black arrow), peripheral hypointensity (arrowhead), and peritumoral retention (white arrow). These patterns should not be confused for uptake in an FNH lesion. The intrahepatic cholangiocarcinoma also shows enhancement of intralesional bile ducts (*).

pulmonary circulation, most commonly involving shunts that directly connect the inferior vena cava with the right pulmonary artery (35). This shunting creates chronically elevated systemic venous pressure, which results in multiorgan venous congestion that can progressively compromise organ function. In the liver, which is particularly vulnerable to this phenomenon, this process has been termed *Fontan-associated liver disease (FALD)* (36). FALD can ultimately progress to cirrhosis and is a substantial non-cardiac contributor to mortality in patients with FALD who survive into adulthood (37).

FNH-like lesions are thought to form in patients with FALD because of hepatic venous hypertension, which diminishes portal flow to the hepatic perivenular space, with resulting arterIALIZATION and responsive hepatocytic hyperplasia (38). FNH-like lesions are very commonly seen with FALD, with a prevalence of 20%–50% of patients (38,39). FNH-like lesions in patients with FALD are typically small (median size, 1 cm), are located peripherally (75% are less than 2 cm from the liver margin), and show APHE (92%), and there are usually multiple lesions (39,40) (Fig 10A). Although the risk is relatively low (1.3%), patients with FALD are at risk for development of HCC (41). Confounding diagnosis at imaging, 7%–10% of FNH-like lesions in patients with FALD may show washout, which is seen more commonly during the delayed phase than during the portal venous phase after contrast material enhancement (39,40,42) (Fig 10B). The possibility of a washout appearance in

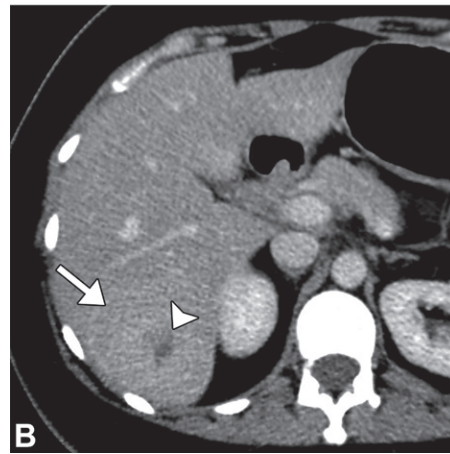
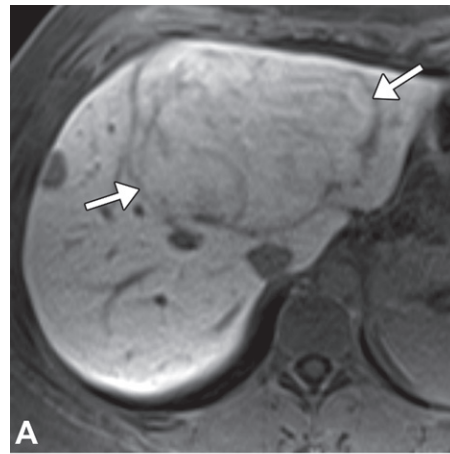


Figure 7. Hepatocellular adenoma (HCA) mimicking FNH. (A) Axial HBP MR image acquired at 20 minutes after administration of gadoxetate in a 21-year-old woman with a β -catenin HCA (arrows) shows isointense to slightly hyperintense uptake of contrast material, which mimics the appearance of FNH. (B) Axial portal venous phase CT image in a 38-year-old woman with a β -catenin HCA (arrow) shows a central scar (arrowhead) mimicking FNH.

FNH-like lesions and their frequency in FALD diminishes the specificity of criteria for imaging diagnosis of HCC in this population. The Liver Imaging Reporting and Data System (LI-RADS) acknowledges this decreased specificity for imaging diagnosis of HCC in populations who are predisposed to formation of FNH-like lesions and explicitly states that LI-RADS criteria should not be applied in patients with cardiogenic cirrhosis (which includes FALD) or with cirrhosis secondary to vascular disorders, including hereditary hemorrhagic telangiectasia (HHT), Budd-Chiari syndrome (BCS), chronic portal vein occlusion, and nodular regenerative hyperplasia (NRH) (43).

Ancillary features of HCC, including T2-hyperintensity, restricted diffusion, and growth on serial MRI examinations, should be considered in conjunction with enhancement patterns. Use

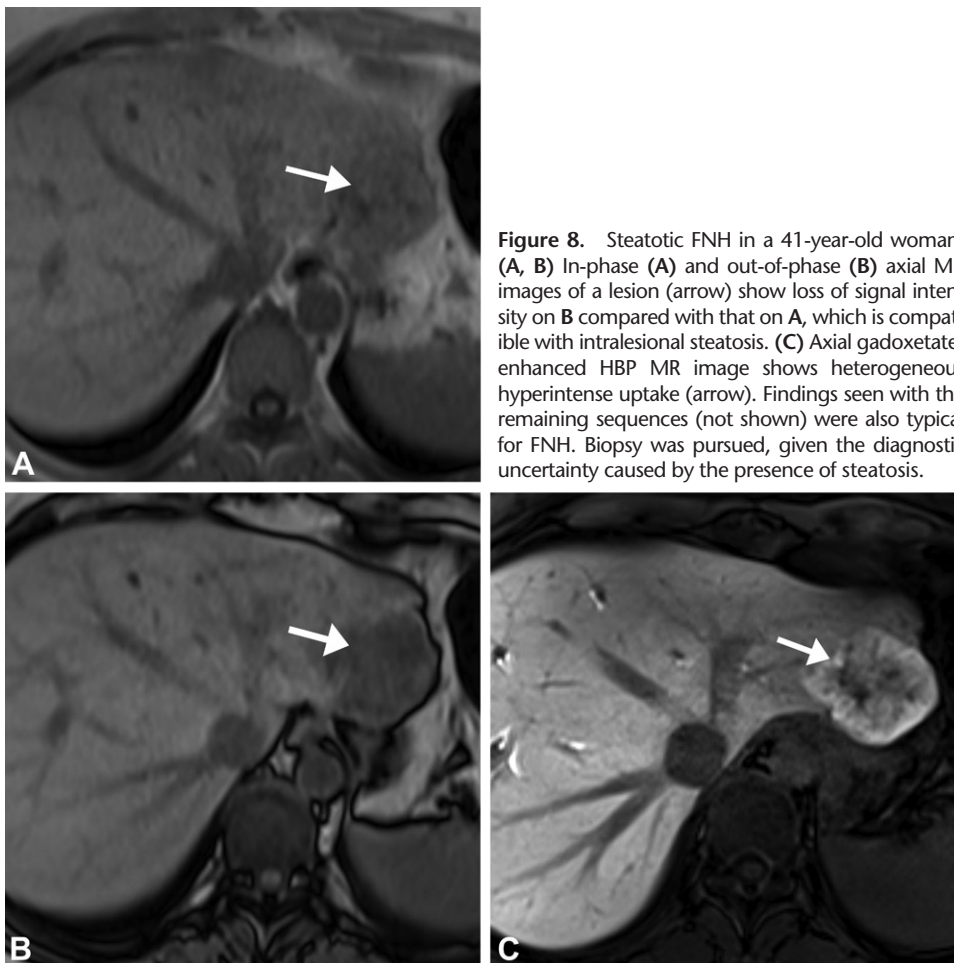


Figure 8. Steatotic FNH in a 41-year-old woman. (A, B) In-phase (A) and out-of-phase (B) axial MR images of a lesion (arrow) show loss of signal intensity on B compared with that on A, which is compatible with intralesional steatosis. (C) Axial gadoxetate-enhanced HBP MR image shows heterogeneous hyperintense uptake (arrow). Findings seen with the remaining sequences (not shown) were also typical for FNH. Biopsy was pursued, given the diagnostic uncertainty caused by the presence of steatosis.

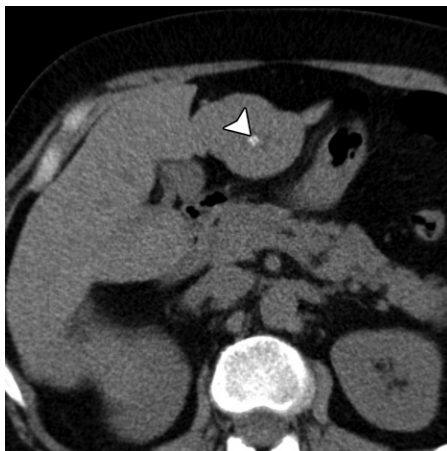


Figure 9. Left hepatic lobe FNH in a 53-year-old woman. Axial unenhanced CT image shows punctate calcification in the central scar (arrowhead). Calcification is rare in FNH and is considered an atypical feature. This lesion showed stability for 10 years and also had features at MRI that were typical of FNH (not shown).

HCC in patients with FALD make the feature of HSCA uptake highly predictive of an FNH-like lesion (Fig 10C). A nodule with typical FNH-like features in a patient with FALD should not cause deviation from routine surveillance. However, a nodule with washout or other atypical features may still be an FNH-like lesion, and characterization with an HSCA or short-interval follow-up imaging should be considered. Ultimately, biopsy may be required to establish the diagnosis of HCC in this population (39,40).

Any condition causing chronic failure of the right side of the heart may lead to chronic hepatic venous congestion and may progress to cardiogenic cirrhosis. Although it is less well studied, congestive hepatopathy from causes other than Fontan circulation also has a propensity for formation of FNH-like lesions, and FNH-like lesions should be a high consideration in the differential diagnosis of any arterially enhancing liver lesion in these patients (44).

of MRI with HSCAs can also be helpful in this population. While a small percentage of HCCs may show HSCA uptake, the frequency of FNH-like lesions and the relatively low incidence of

Budd-Chiari Syndrome.—The primary form of BCS occurs from obstruction of hepatic venous outflow, usually because of hepatic vein thrombosis; intrinsic venous narrowing; or more rarely,

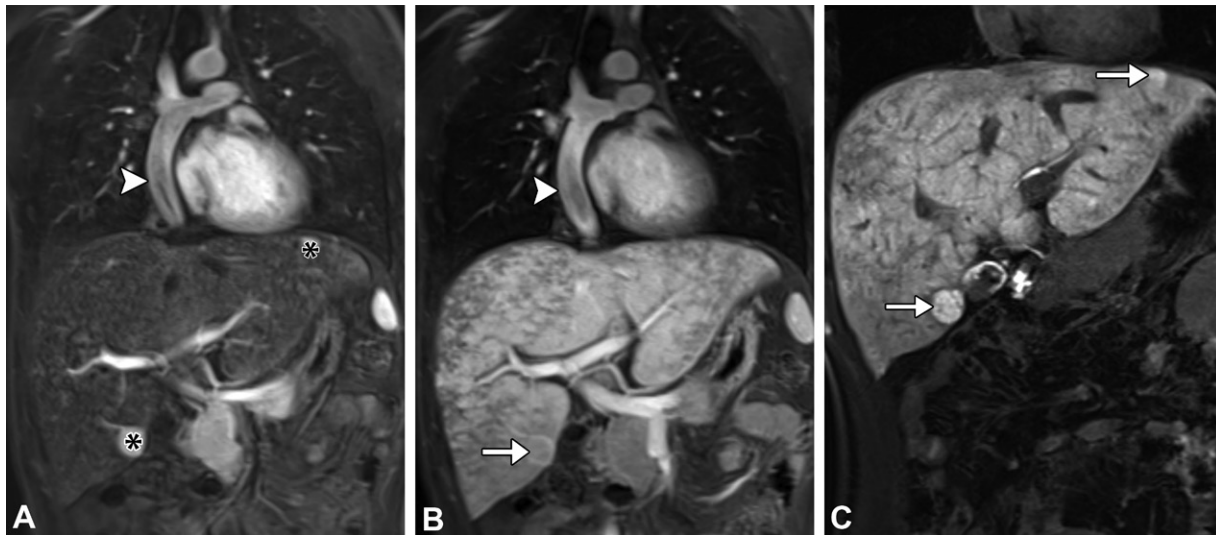


Figure 10. Images of the heart and liver in a 26-year-old man after he underwent Fontan surgery. (A, B) Coronal arterial phase (A) and portal venous phase (B) gadobutrol-enhanced MR images show two (of multiple) APHE liver lesions (* in A), one of which demonstrates washout appearance (arrow in B). (C) Subsequent coronal gadoxetate-enhanced HBP MR image shows homogeneously hyperintense uptake that is consistent with FNH-like lesions (arrows). Note the widely patent Fontan shunt (arrowhead in A and B).

webs obstructing the intrahepatic inferior vena cava or hepatic veins. The secondary form can occur from extrinsic compression or direct venous invasion by the tumor. Outflow obstruction leads to hepatic congestion and increased sinusoidal pressure, with development of portal hypertension and parenchymal fibrosis. Intra- and extrahepatic venous collateralization is common, and responsive hyperarterialization and hypertrophy in areas of preserved hepatic venous outflow predispose patients to formation of FNH-like lesions (45). Previously, these lesions in patients with BCS have been variably described as adenomatous hyperplastic nodules, large regenerative nodules, and FNH-like regenerative nodules (46). However, the term *FNH-like lesion* is most accurate and consistent because they represent the same process as that in patients with other hepatic vascular disturbances.

FNH-like lesions are reported in more than one-third of patients with chronic BCS (47). There are often multiple lesions that can increase or decrease in size and number over time. Unlike typical FNH, most FNH-like lesions in patients with BCS have T1-hyperintense and T2-hypointense signal relative to the background liver at MRI (47,48) (Fig 11). Other typical features are usually present, including APHE and uptake of HSCA. Subtraction imaging increases the conspicuity of APHE, given the frequent T1-hyperintensity seen at MRI (46,48).

Patients with BCS are at increased risk for HCC. Up to one-third of FNH-like lesions in patients with BCS show washout, emphasizing the lack of specificity of classic imaging criteria for the diagnosis of HCC in this population (48).

It is important to assess arterially enhancing lesions in patients with BCS for features supportive of diagnosis of an FNH-like lesion, including T1-hyperintensity, lack of diffusion restriction, and the presence of a central scar at MRI; normal α -fetoprotein levels; and lack of rapid growth. MRI with an HSCA is helpful and, in combination with other imaging features, can allow FNH-like lesions to be distinguished from HCCs with confidence in most cases (46).

Abnormalities of Hepatic Inflow

Congenital Absence of the Portal Vein.—*Congenital absence of the portal vein*, or *Abernethy malformation*, is a rare malformation resulting in bypass of splanchnic venous flow from the liver directly to the inferior vena cava. Congenital absence of the portal vein is subtyped on the basis of the congenital extrahepatic portosystemic shunt that is present, with the most common type being the complete absence of the portal vein and drainage of the splenic and superior mesenteric veins directly into a systemic vein separately or through a short common trunk (49).

Although congenital absence of the portal vein is rare, FNH-like lesions are frequently reported in this population (Fig E6). In a review of 101 cases of congenital absence of the portal vein, hepatic masses were seen in almost one-half of patients. Of these, the majority were FNH-like lesions (36.7%), followed in frequency by HCCs (26.5%), NRHs (16.3%), HCAs (10.2%), and others (50). The absence of portal blood flow with compensatory increased hepatic arterial blood flow has been theorized as the cause of the

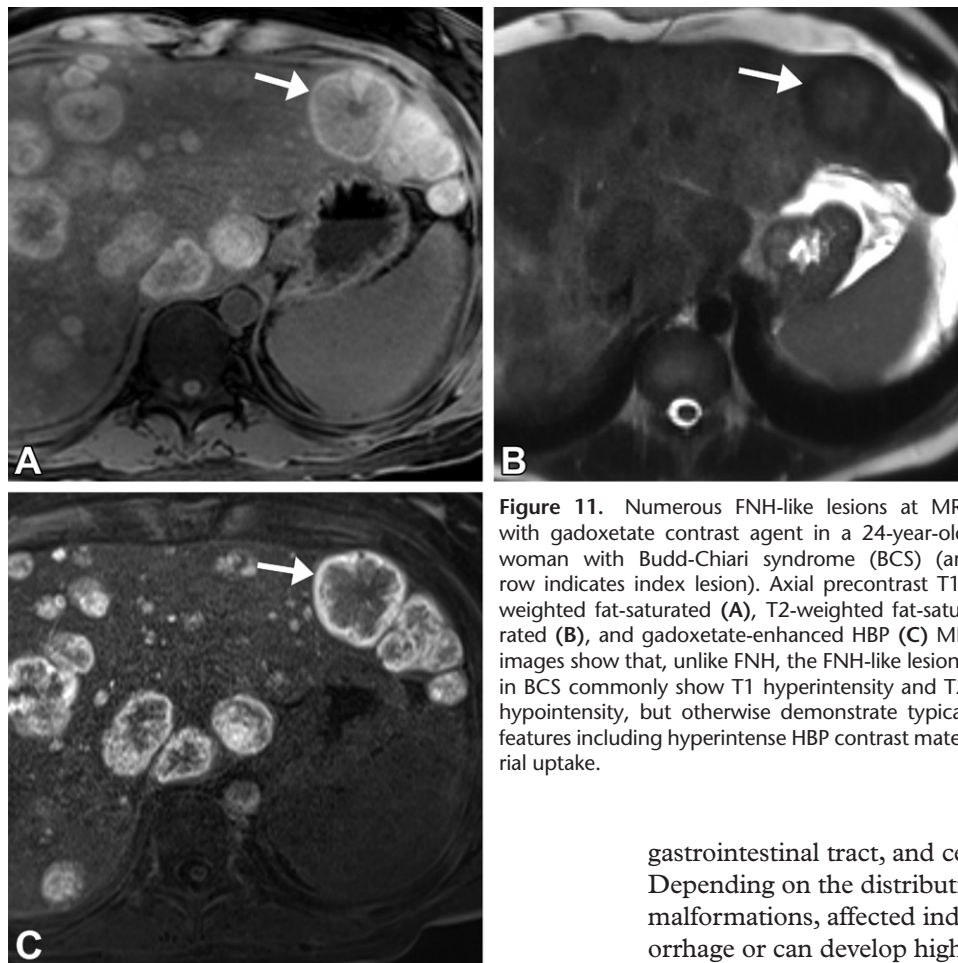


Figure 11. Numerous FNH-like lesions at MRI with gadoxetate contrast agent in a 24-year-old woman with Budd-Chiari syndrome (BCS) (arrow indicates index lesion). Axial precontrast T1-weighted fat-saturated (A), T2-weighted fat-saturated (B), and gadoxetate-enhanced HBP (C) MR images show that, unlike FNH, the FNH-like lesions in BCS commonly show T1 hyperintensity and T2 hypointensity, but otherwise demonstrate typical features including hyperintense HBP contrast material uptake.

frequency of FNH-like lesions and other hepatic masses in these patients (51).

Cavernous Transformation.—*Cavernous transformation* refers to the development of numerous collateral veins in the hepatic hilum as a response to restoring flow after portal vein thrombosis. This substantial alteration in hepatic hemodynamics leads to increased arterial perfusion and an accompanying predilection for formation of FNH-like lesions. In a study of 58 patients with cavernous transformation, 21% were found to have at least one FNH-like lesion. In this setting, imaging features are typical and are like those in the other predisposing conditions we have described, including small size, peripheral location, and homogeneous APHE (52).

Hereditary Hemorrhagic Telangiectasia.—HHT, also known as Osler-Weber-Rendu syndrome, is an autosomal-dominant disorder that is characterized by telangiectasia, or abnormal communication between an arteriole and a dilated postcapillary venule. Arteriovenous malformations in patients with HHT increase with age and can involve the skin, mucous membranes, lungs, liver,

gastrointestinal tract, and central nervous system. Depending on the distribution of arteriovenous malformations, affected individuals can have hemorrhage or can develop high-output cardiac failure from hepatic shunting. Diagnosis of HHT requires three of four findings: (a) recurrent epistaxis since childhood, (b) cutaneous or mucosal telangiectases, (c) visceral involvement, and (d) family history of HHT (53).

The liver is the most commonly involved abdominopelvic organ in patients with HHT, and liver involvement in HHT often develops during adulthood. Although most patients are asymptomatic, with preserved liver function, up to 8% become symptomatic as a result of the *arterial steal phenomenon*, from arterioportal or arteriovenous shunts progressing to ischemic cholangiopathy, increased portal flow from shunting evolving into portal hypertension, or portovenous shunts resulting in high-output cardiac failure (53).

Hepatic imaging findings of HHT include enlarged hepatic arteries (diameter, ≥ 6 mm), intrahepatic arteriovenous or portovenous shunts, perfusion anomalies, telangiectases, large confluent vascular masses, and FNH-like lesions (54–57) (Fig 12). While some literature describes the large confluent vascular masses of HHT separately from FNH-like lesions, studies (56,57) with specific imaging criteria for FNH-like lesions or histologic evaluation support that these masses are also FNH-like lesions. Patients with HHT often

have multiple FNH-like lesions, with a reported prevalence as high as 47% in those with liver involvement (54,55,58). In our experience with MRI with HSCAs, many small FNH-like lesions confirmed with hyperintense HBP uptake have an appearance indistinguishable from telangiectases and large confluent vascular masses at CT or MRI with extracellular contrast agents (Fig 13).

Because a pseudocirrhotic appearance of the liver is common in patients with HHT, APHE lesions may be considered suspicious for HCCs by those unfamiliar with this condition. However, HCC in patients with HHT is rare, and these patients are not thought to be at increased risk for development of primary hepatic malignancies (59). Most FNH-like lesions in patients with HHT have a typical appearance, although they may lack a central scar or have heterogeneous or absent APHE because of arterial shunting (58). Given the frequency of FNH-like lesions and the relative rarity of HCC in patients with HHT, a mass with atypical features is still likely to be an FNH-like lesion. Because of the increased risk of bleeding in these patients, percutaneous biopsy should be avoided whenever possible. MRI with HSCAs or imaging follow-up should be the preferred tool for assessment and diagnosis of a hepatic mass in a patient with HHT.

Hepatic Microvascular Disturbances

Cirrhosis.—*Cirrhosis of the liver* is the pathologic result of chronic liver damage with necroinflammation and progressive fibrosis (60). Accurate imaging evaluation is of the utmost importance in patients with cirrhosis because of the substantially increased risk of malignancy, most commonly HCC. Because HCC is usually diagnosed on the basis of imaging, without the need for pathologic confirmation, radiologists must be well informed regarding its imaging appearance and those of its mimics, including FNH-like lesions, which have been reported in 3%–15% of patients with cirrhotic livers (19,61).

FNH-like lesions are thought to form because of the altered vascular dynamics of cirrhosis, including sinusoidal remodeling, local ischemia and angiogenesis, and formation of intrahepatic shunts (34,60). FNH-like lesions, unlike dysplastic nodules, do not undergo malignant transformation and are therefore considered “do not touch” lesions (34,42,61). Histologically, they can be challenging to distinguish from well-differentiated HCCs, with one study (42) reporting a 6.5% rate of misdiagnosis as HCC at percutaneous core needle biopsy.

Diagnostic imaging features of HCC in patients with cirrhosis include APHE, venous



Figure 12. Hereditary hemorrhagic telangiectasia (HHT) in a 48-year-old woman. Coronal venous phase gadobutrol-enhanced MR image shows enlarged hepatic arteries (arrowheads), which are characteristic of liver involvement in HHT. An isoenhancing FNH-like lesion (arrow) is a common finding in these patients.

washout, and a pseudocapsule (43). FNH-like lesions can be difficult to distinguish from HCC in the cirrhotic liver, given the shared feature of APHE, especially when they measure smaller than 3 cm. Although venous washout and a pseudocapsule are characteristic of HCC, these features can be seen in a minority (6%–10%) of FNH-like lesions. Choi et al (42) found that three of nine (33%) FNH-like lesions would have been misdiagnosed as HCC at imaging if the usual diagnostic criteria were applied.

With the use of HSCAs, FNH-like lesions in patients with cirrhosis typically demonstrate isointensity or hyperintensity at HBP MRI (Fig 14). However, up to 30% of FNH-like lesions in patients with cirrhosis may demonstrate signal hypointensity at HBP MRI (62). Confusing the issue further, 6%–15% of HCCs may overexpress hepatocyte-specific membrane transport proteins, with resultant isointense or hyperintense HSCA uptake at HBP MRI (15) (Fig 15). So how can these lesions be differentiated in the cirrhotic liver? Ancillary features, including T2 hyperintensity and increased signal intensity at diffusion-weighted MRI, can support the diagnosis of HCC (19). HCCs typically lack the characteristic scar that can be seen with an FNH-like lesion. A study (63) evaluating HCC and FNH-like lesions with HSCA uptake at HBP MRI showed that washout during the venous phase was the most accurate predictor of

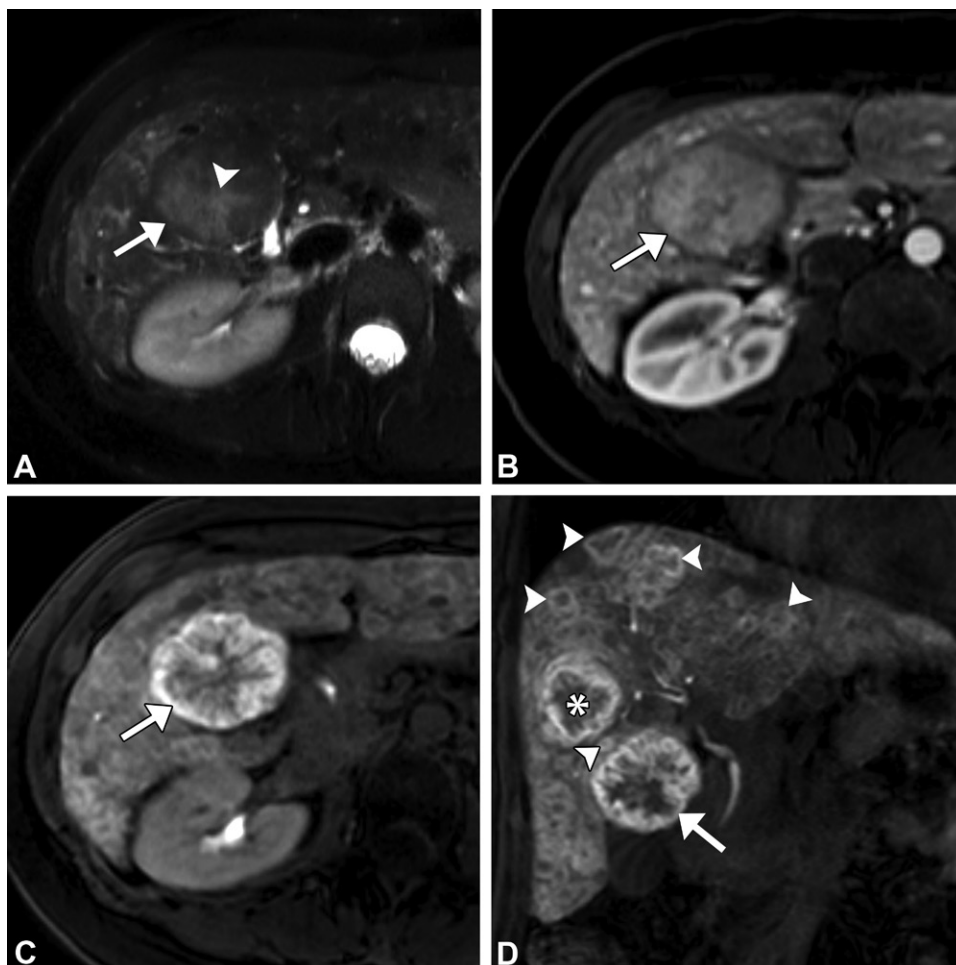


Figure 13. HHT and FNH-like lesions (arrow) with typical features at MRI in a 44-year-old woman. (A) Axial precontrast T2-weighted fat-saturated MR image shows characteristic features of an FNH-like lesion, with a T2-hyperintense central scar (arrowhead). (B) Axial gadoxetate-enhanced arterial phase MR image shows homogeneous APHE. (C, D) Axial (C) and coronal (D) gadoxetate-enhanced HBP MR images show homogeneous hyperintense uptake. The coronal image (D) shows multiplicity, with an additional FNH-like lesion (*) and numerous smaller FNH-like lesions (arrowheads) that were indiscernible from the background of telangiectases and “large confluent vascular masses” that were described in reports from a prior MRI examination (not shown) performed with an extracellular contrast agent.

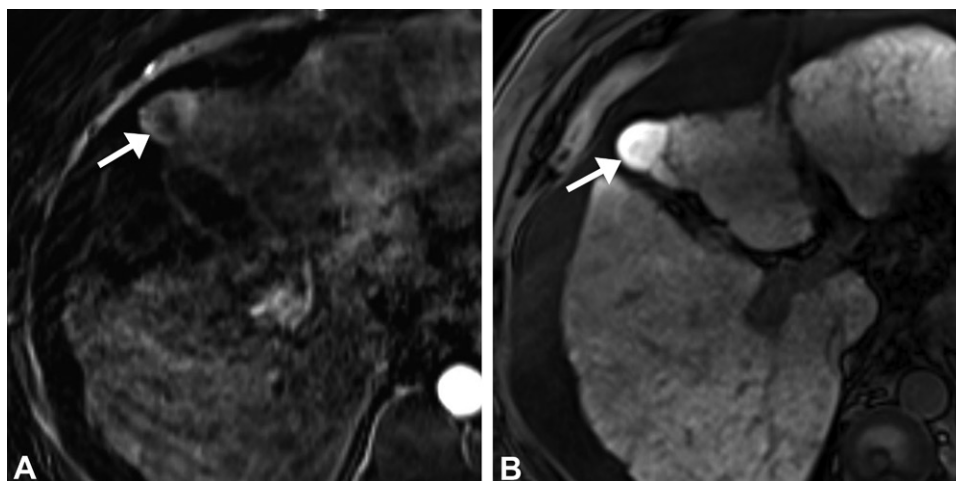


Figure 14. Pathologically proven FNH-like lesion in a 56-year-old man with cirrhosis. (A) Axial gadoxetate-enhanced arterial phase subtraction MR image shows APHE (arrow). (B) Axial gadoxetate-enhanced HBP MR image shows typical homogeneous hyperintense contrast material uptake (arrow).

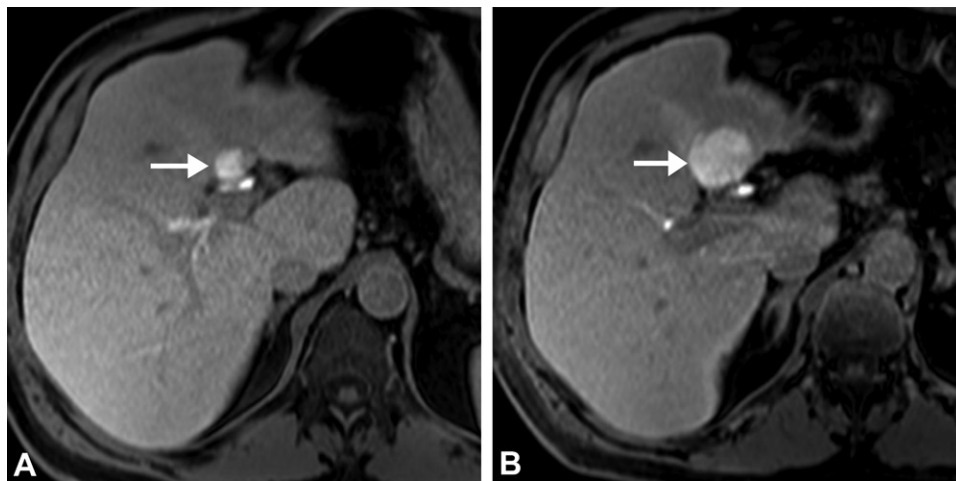


Figure 15. Pathologically proven HCC in a 68-year-old man with cirrhosis. (A) Axial gadoxetate-enhanced HBP MR image shows the lesion (arrow) with homogeneous hyperintense uptake at baseline. (B) Axial gadoxetate-enhanced HBP MR image at 6-month follow-up shows growth of the lesion (arrow), suggesting the correct diagnosis of HCC.

HCC, with 90% sensitivity and 100% specificity in a small sample of lesions. Similarly, another study (62) showed that small size (<1.6 cm), signal hypointensity or isointensity at diffusion-weighted MRI, and lack of portal venous wash-out were independent predictors of FNH-like lesions, and that if all three criteria were met, then 100% specificity could be achieved. In this study, signal intensity during the HBP was not an independent factor for distinguishing FNH-like lesions from HCCs. Although gadoxetate may not be ideal for screening examinations in all patients with cirrhosis because of increased imaging time, artifacts related to transient tachypnea, and impaired HSCA uptake in patients with severely diminished hepatic function, the additional information provided by HBP MRI may help in difficult cases and may be advantageous in patients with cirrhosis secondary to one of the hepatic vascular disturbances we have described, given their increased tendency to develop FNH-like lesions.

Given the potential imaging overlap and the prevalence of HCC in patients with cirrhosis, LI-RADS advises against classifying lesions thought to reflect FNH-like lesions as LR-1 (benign), suggests that LR-2 (probably benign) be used with caution, and recommends that these lesions be categorized as LR-3 (intermediate probability) (43). For patients with cirrhosis for whom LI-RADS cannot be applied, a similar conservative approach of close imaging surveillance of a lesion thought to be FNH-like seems warranted.

Nodular Regenerative Hyperplasia.—NRH is a rare hepatic vascular disorder characterized by diffuse benign transformation of the hepatic parenchyma into small (<3 mm) regenerative

Table 3: Conditions Associated with NRH

Hematologic
Lymphoproliferative disorders
Myeloproliferative disorders
Common variable immunodeficiency syndrome
HIV infection
Autoimmune
Inflammatory bowel disease
Rheumatoid arthritis
Systemic lupus erythematosus
Vascular
Congenital absence of the portal vein
Portal vein thrombosis
Budd-Chiari syndrome
Medications
Chemotherapeutics
Immunosuppressants
Antivirals

nodules that may sometimes coalesce to form larger macronodules. In comparison to cirrhosis, minimal to no intervening fibrosis is seen. Although the incidence of NRH at autopsy is only 0.6%–2.6%, it is the leading cause of noncirrhotic intrahepatic portal hypertension (64).

The exact etiopathogenesis of NRH remains unclear. One factor contributing to its development is microvascular insult to small hepatic veins giving rise to obliterative portal venopathy. This leads to hepatic parenchymal atrophy from underlying hepatocytic ischemia, with resultant compensatory hepatic nodular regeneration (65). NRH has been linked to a host of different diseases, including various hematologic, autoimmune, and vascular disorders, and to several medications (Table 3).

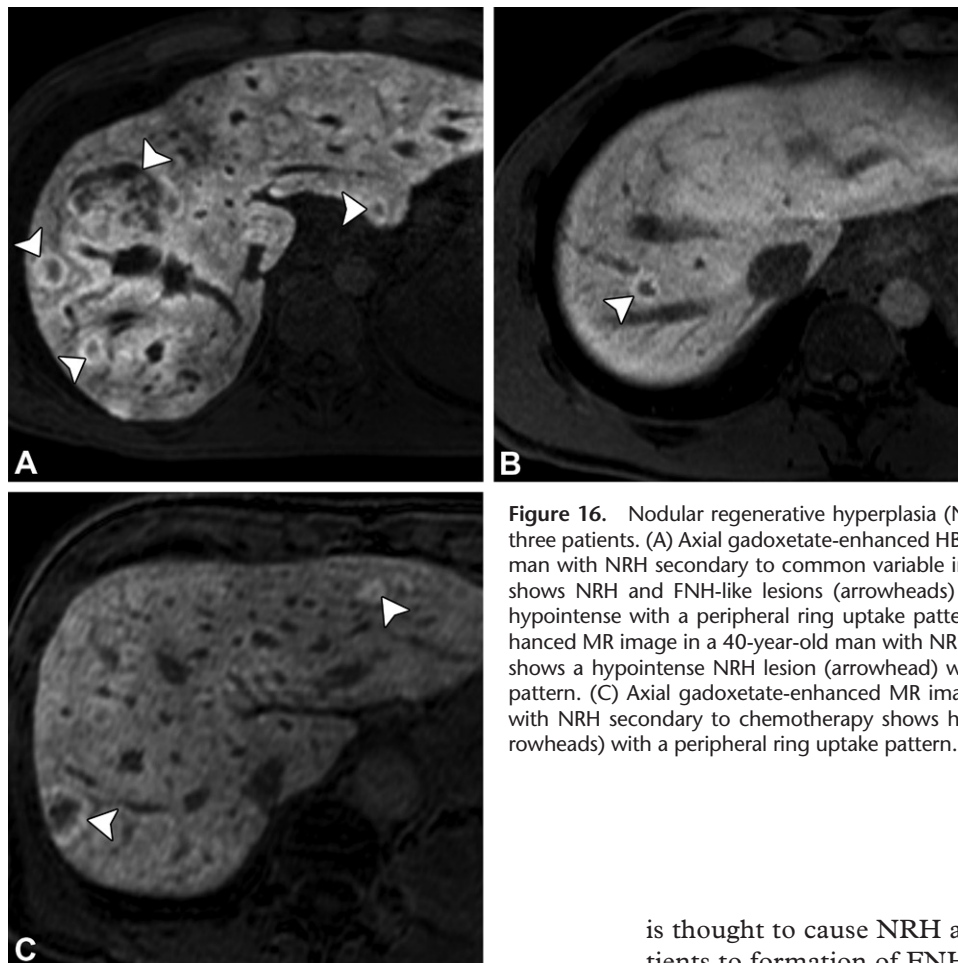


Figure 16. Nodular regenerative hyperplasia (NRH) and FNH-like lesions in three patients. (A) Axial gadoxetate-enhanced HBP MR image in a 44-year-old man with NRH secondary to common variable immunodeficiency syndrome shows NRH and FNH-like lesions (arrowheads) that appear predominantly hypointense with a peripheral ring uptake pattern. (B) Axial gadoxetate-enhanced MR image in a 40-year-old man with NRH secondary to HIV infection shows a hypointense NRH lesion (arrowhead) with a peripheral ring uptake pattern. (C) Axial gadoxetate-enhanced MR image in a 65-year-old woman with NRH secondary to chemotherapy shows hypointense NRH lesions (arrowheads) with a peripheral ring uptake pattern.

NRH is often a diagnostic dilemma, because most patients are asymptomatic at initial presentation, with normal laboratory findings. When symptoms are present, they are usually related to the underlying systemic disease or portal hypertension. Biopsy is considered the standard of care for diagnosis of NRH, because the imaging features are neither specific nor sensitive (66).

Most patients, especially those without portal hypertension, have a normal-appearing homogeneous liver at imaging, because the 1–3-mm nodules typical of NRH are too small to discern. However, some patients may have larger nodules (up to 15 mm) that are identifiable with imaging. Literature regarding the presence of larger nodules that are readily identified with imaging in patients with NRH is confusing. Although one study (67) of 36 patients with NRH demonstrated no enhancing liver masses, multiple other studies (15,19,68–70) report enhancing masses that have variably been termed *large regenerative nodules* but are very likely FNH-like lesions. NRH and FNH-like lesions are described as separate entities, although they coexist in some patients in clinical practice (Fig 16). It may be that the underlying vascular insult that

is thought to cause NRH also predisposes patients to formation of FNH-like lesions or that this coappearance is related to an overlap in the underlying populations that are predisposed to both conditions, such as in those with a congenital absence of the portal vein or with exposure to chemotherapy. These lesions in patients with NRH may (but are less likely to) demonstrate arterial hyperenhancement and are more likely to demonstrate mild to moderate hyperenhancement at venous phase MRI than are typical FNH lesions (Fig E7). On MR images, FNH-like lesions observed in patients with NRH tend to show hypointensity with a peripheral ring uptake pattern during the HBP (15) (Fig 16). A link between NRH and development of HCC has been suggested (71). MRI with HSCAs may be helpful to characterize masses found at screening, because HSCA uptake during the HBP is a strong predictor of benignity regardless of the nomenclature used for the benign nodules discovered in these patients.

Chemotherapy-induced FNH-like Lesions.—Toxicity from chemotherapy can induce a variety of hepatic changes, including acute hepatocellular injury, steatosis and steatohepatitis, sinusoidal obstruction syndrome, and NRH (72). In addition to diffuse parenchymal changes, formation of FNH-like lesions is increasingly recognized as a

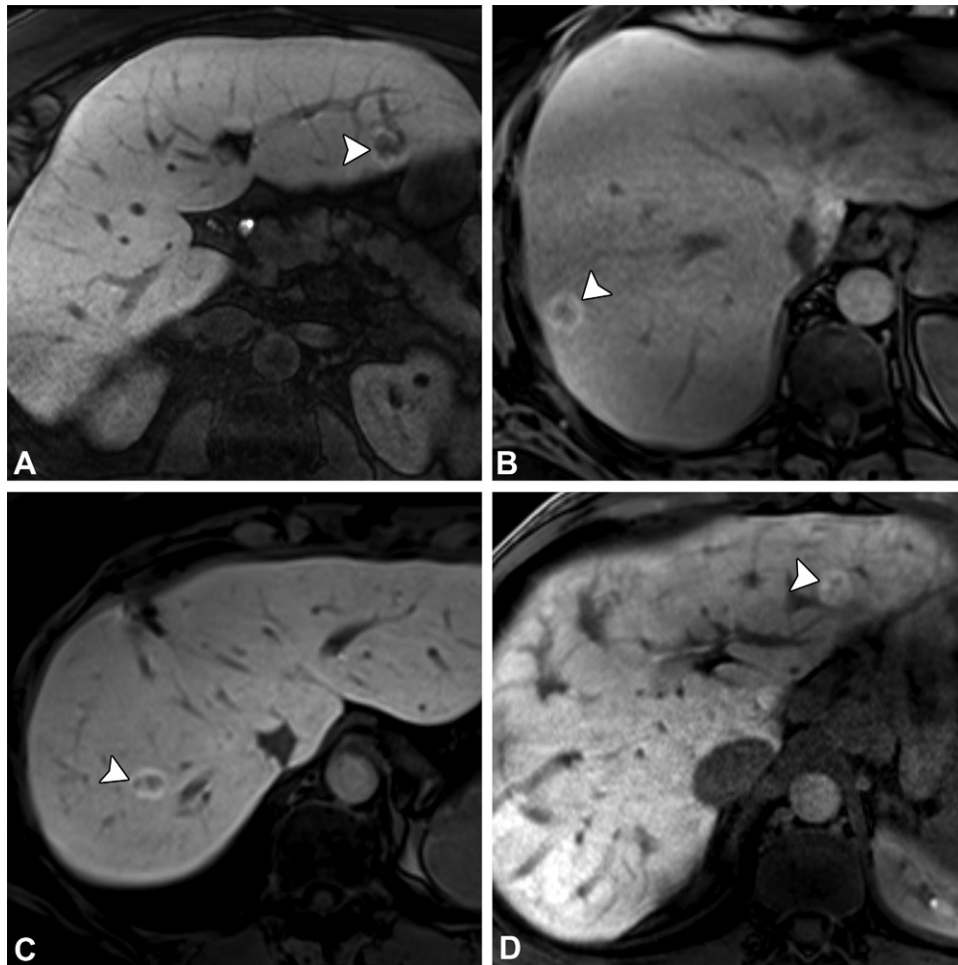


Figure 17. New FNH-like lesions in four patients after treatment with oxaliplatin-based chemotherapy. (A–C) Axial gadoxetate-enhanced HBP MR images in a 60-year-old woman who was treated for colon adenocarcinoma (A), a 49-year-old man who was treated for rectal adenocarcinoma (B), and a 62-year-old woman who was treated for colon adenocarcinoma (C) show a hypointense FNH-like lesion (arrowhead) with a peripheral rim uptake pattern. (D) Axial gadoxetate-enhanced HBP MR image in a 38-year-old man who was treated for colon adenocarcinoma shows an FNH-like lesion (arrowhead) with a homogeneously hyperintense uptake pattern.

consequence of chemotherapy. Particularly, adult treatment with oxaliplatin (a common component of regimens to treat colorectal and pancreatic adenocarcinomas) and cisplatin and other agents in childhood have most commonly been linked with posttreatment development of FNH-like lesions (73,74). The exact cause is uncertain, but they are hypothesized to be secondary to chemotherapy-related vascular injury (72).

New and growing FNH-like lesions in patients undergoing surveillance for cancers that are prone to metastasize to the liver are problematic and can lead to unnecessary invasive procedures if they are mistaken for metastases. In a study of 14 patients with FNH-like lesions that developed after they underwent oxaliplatin chemotherapy, the majority of patients (86%) had multiple FNH-like lesions that were small (average size, approximately 2 cm) and occurred over a wide range of 10–120 months (average, 47.6 months) after treatment

(73). These lesions had typical FNH-like features, including APHE and hyperintense or isointense uptake of HSCA during the HBP at MRI (73,74) (Fig 17). Furlan et al (73) found that FNH-like lesions after oxaliplatin therapy most frequently showed HBP hypointensity, with a peripheral ring uptake pattern at MRI. Central scars were infrequently described, although this may be related to the small average size of the lesions. The size and number of FNH-like lesions may increase with time (73,74) (Fig 18).

Metastases from the adult malignancies that are commonly treated with oxaliplatin regimens are most commonly hypoenhancing at MRI. Given the possibility of development of FNH-like lesions in these patients, the determination that a new arterially enhancing lesion at follow-up imaging is a metastasis should be made with caution. HSCA-enhanced MRI is reliable for differentiating metastatic and FNH-like lesions

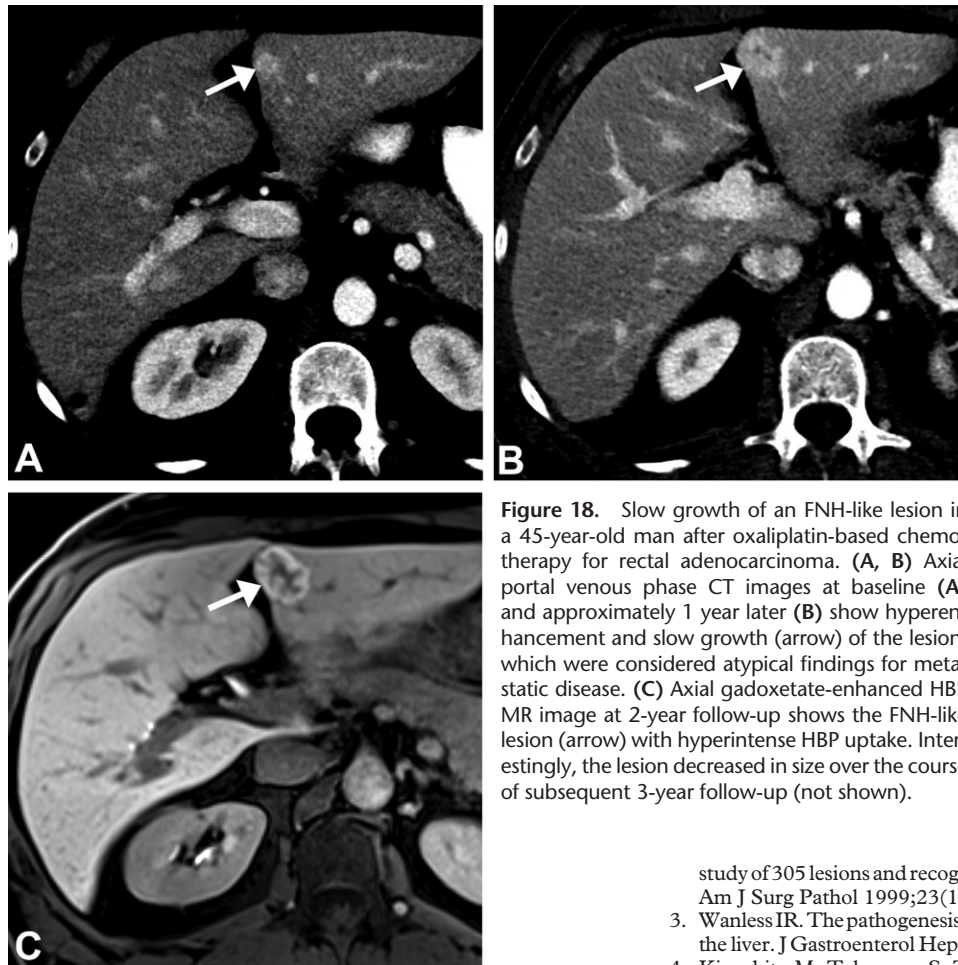


Figure 18. Slow growth of an FNH-like lesion in a 45-year-old man after oxaliplatin-based chemotherapy for rectal adenocarcinoma. (A, B) Axial portal venous phase CT images at baseline (A) and approximately 1 year later (B) show hyperenhancement and slow growth (arrow) of the lesion, which were considered atypical findings for metastatic disease. (C) Axial gadoxetate-enhanced HBP MR image at 2-year follow-up shows the FNH-like lesion (arrow) with hyperintense HBP uptake. Interestingly, the lesion decreased in size over the course of subsequent 3-year follow-up (not shown).

and should be used when possible. If the patient is not a candidate for MRI, contrast-enhanced US or serial imaging should be considered as an alternative. Slow growth or the appearance of additional similar lesions should not be mistaken for a suspicious finding if these lesions demonstrate features typical of FNH-like lesions, because this is likely part of the natural course of this phenomenon.

Conclusion

FNH and FNH-like lesions are benign lesions that can often be diagnosed on the basis of imaging. Knowledge of the imaging features of FNH and of the numerous conditions that predispose a patient to the formation of FNH-like lesions and their characteristic imaging appearance can help radiologists to guide appropriate conservative management of these benign, clinically silent liver lesions.

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